Week 4-1

Discrete Random Variable

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2110205 STATS COMP ENG

CHULA
COMPUTER
ΣNGINEERING

Probability Recap

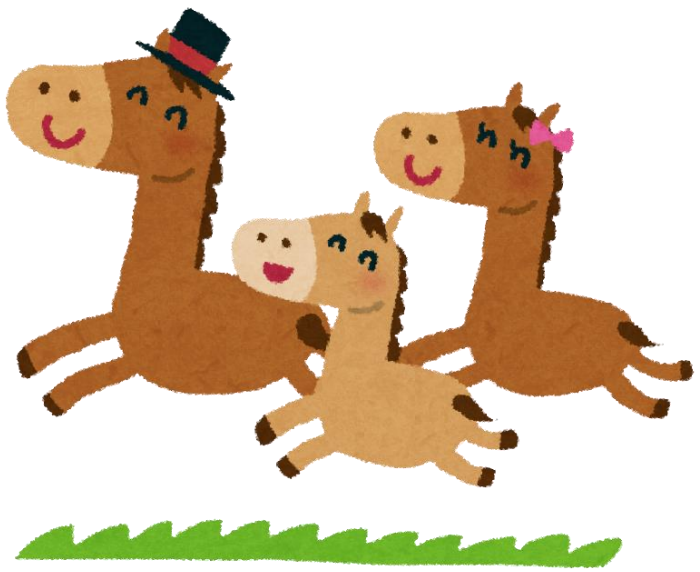
Probability Recap

- Discrete sample space $\Omega = \{\omega_1, \omega_2, \dots\}$
  - Probability of ω_i is $P(\omega_i)$
- Event $E \subset \Omega$
- Probability of E is $P(E) = \sum_{\omega \in E} P(\omega)$

Example

- We race 3 horses: A, B, C
- The probability of:

ABC	30%	**ACB**	20%
BAC	20%	**BCA**	10%
CAB	15%	**CBA**	5%



- We have $\Omega = \{ABC, ACB, BAC, BCA, CAB, CBA\}$
- Event A wins = $\{ABC, ACB\} \subset \Omega$
- P(A wins) = P(ABC)+P(ACB) = 0.5

Let make a bet

Horse Racing

Given these returns, we can write a **<u>payoff function</u>**:

$$X = \begin{cases} 100 & if\ A\ wins \\ 150 & if\ B\ wins \\ 200 & if\ C\ wins \end{cases}$$

Who wins	Payout
A	100
B	150
C	200

Who should we bet on? We will come back later.

Discrete Random Variable

Definition: Discrete Random Variable

A function that assign **value** (R) to each **outcome** (ω).

<u>Example</u>: $X = \begin{cases} \boxed{100} & if \ \boxed{A \ wins} \\ \boxed{150} & if \ \boxed{B \ wins} \\ \boxed{200} & if \ \boxed{C \ wins} \end{cases}$

Attendance Check

Problem: We tossed a fair coin 1 time. Is the following X a discrete random variable when 0 means HEAD and 1 mean TAIL?

$$X(0) = 0.5$$
$$X(1) = 0.5$$

Answer:

Event and Random Variable

- Back to the horse race.

$$X = \begin{cases} 100 & if\ A\ wins \\ 150 & if\ B\ wins \\ 200 & if\ C\ wins \end{cases}$$

- $X = a$ denotes all outcomes that $X(\omega) = a$.
- $P(X = a)$ is the probability that X evaluate to a.

  - Event $X = 100$ is the set $\{ABC, ACB\}$ P(X=100) = P(ABC)+P(ACB) = 0.5
  - Event $X = 150$ is the set $\{BAC, BCA\}$ P(X=150) = P(BAC)+P(BCA) = 0.3
  - Event $X = 500$ is the set ϕ P(X=500) = 0

Probability Mass Function

- **Probability Mass Function (pmf)** is a function for $P(X = a)$
- That is, pmf $p(a) \ = \ P(X = a)$

- Properties:
  - $0 \leq p(a) \leq 1$
  - $\sum p(a) = 1$
  - a can be any value.

Example

- **We toss 8 fair coins.**

  Let H denote the number of heads, and T denote the number of tail.

- Random variable $\mathrm{Y}(H,T) = |H - T|$.
  - Example: if we get 6 heads and 2 tails, then $Y(6,2) = 4$.

pmf =

Y	0	2	4	6	8
p(a)	**70**/256	**112**/256	**56**/256	**16**/256	**2**/256
	4H4T	5H3T	6H2T	7H1T	8H
		3H5T	2H6T	1H7T	8T

Think: How did we come up with these number?

Example (cont'd)

pmf =

Y	0	2	4	6	8
p(a)	**70**/256	**112**/256	**56**/256	**16**/256	**2**/256
	4H4T	5H3T	6H2T	7H1T	8H
		3H5T	2H6T	1H7T	8T

Small quiz:

- $p(a = 0) =$
- $p(a = 5) =$
- $p(a \leq 5) =$

Cumulative Distribution Function

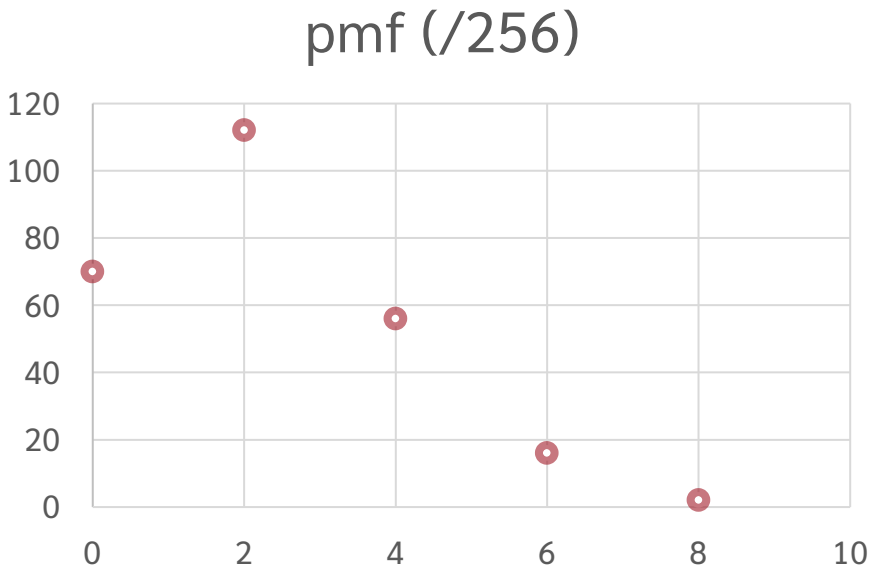
- **Cumulative Distribution Function (cdf)** F for X is

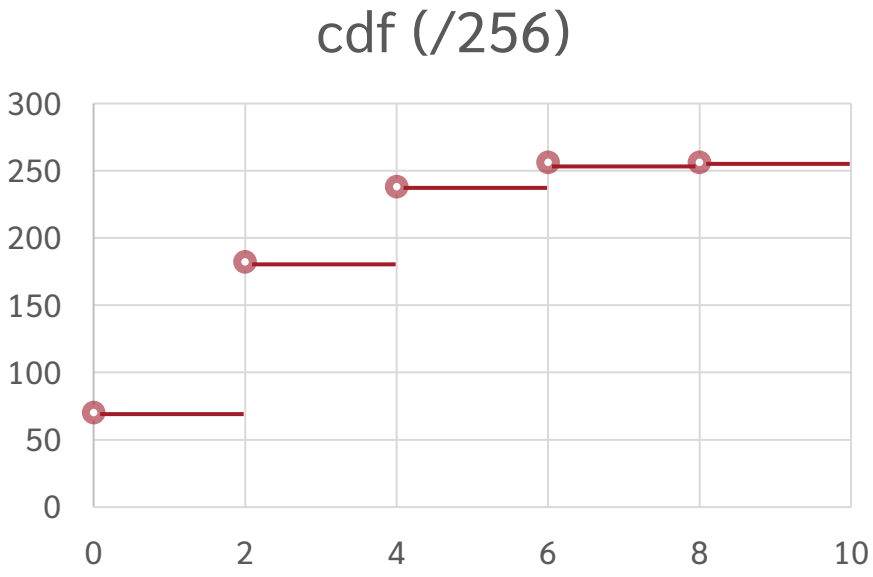
$$F(a) = P(X \leq a)$$

Less than or **EQUAL**

Example

	Y	0	2	4	6	8
pmf =	p(a)	**70**/256	**112**/256	**56**/256	**16**/256	**2**/256
cdf =	F(a)	**70**/256	**182**/256	**238**/256	**254**/256	**256**/256





Quiz 1

	a	4	7	13	23	24
pmf =	p(a)	**1/8**	**2/8**	**1/8**	**???**	**1/8**

1. What is pmf p(23)
2. What is pmf p(25)
3. What is cdf F(2)
4. What is cdf F(24)

10 minutes break

Various kind of distributions

- Uniform distribution
- Bernoulli distribution
- Binomial distribution
- Geometric distribution

- And more…

Uniform distribution

$$X \sim Uniform(N)$$

All N outcomes are <u>equally likely</u>.

<u>Examples</u>: fair coin ($N = 2$), dice ($N = 6$), etc.

Value a	0	1	2	…	N-1
pmf $p(a)$	$1/n$	$1/n$	$1/n$	…	$1/n$
cdf $F(a)$	$1/n$	$2/n$	$3/n$	…	1

Bernoulli distribution

$$X \sim Bernoulli(p)$$

Either *success* or *failure*, where the chance of success is p.

Value a	0	1
pmf $p(a)$	$1-p$	p
cdf $F(a)$	$1-p$	1

Examples:

- Fair coin ($p = 0.5$)
- Vote for acceptance ($p =?$)
- Passing this class ($p = 0.9?$)

Binomial distributions

$$X \sim Binomial(n, p)$$

- If you do something $\boldsymbol{n}$ **times**, each time has **chance of success** $\boldsymbol{p}$**.** How many time would you succeed?
- Or, in another word, you do Bernoulli(p) n times.

Binomial Example

Example: Toss 4 fair coins.

Supposed X follows Binomial distribution with $n = 4$ and $p = 0.5$.

X = 0	X = 1	X = 2	X = 3	X = 4
TTTT	HTTT THTT	HHTT HTHT HTTH	HHHT HHTH	HHHH
	TTHT TTTH	THHT THTH TTHH	HTHH THHH	
$\binom{4}{0} = 1$	$\binom{4}{1} = 4$	$\binom{4}{2} = 6$	$\binom{4}{3} = 4$	$\binom{4}{4} = 1$

Prob = $\binom{4}{0}(1-p)^4p^0$ $\binom{4}{1}(1-p)^3p^1$ $\binom{4}{2}(1-p)^2p^2$ $\binom{4}{3}(1-p)^1p^3$ $\binom{4}{4}(1-p)^0p^1$

Binomial distribution pmf

$$X \sim Binomial(n, p)$$

$$p(a) = \binom{n}{a}(1-p)^{n-a}p^{a}$$

Geometric distribution

$$X \sim Geometric(p)$$

How many FAILURE we get until we get a SUCCESS?
(Success probability p)

Geometric example

Example: Suppose the chance you pass this course is $p = 0.9$.

Let X be the number of times you failed the course before passing.

- Pass on 1st try: $p = 0.9$
- Pass on 2nd try: $p = 0.1 \times 0.9$
- Pass on 3rd try: $p = 0.1^2 \times 0.9$
- And so on…

Value a	0	1	2	…	k
pmf $p(a)$	p	$(1-p)p$	$(1-p)^2 p$	…	$(1-p)^k p$

Distribution summary

	Parameter	Domain of a	pmf $p(a)$
Uniform(n)	n number of possible outcomes	1 to n	$\frac{1}{n}$
Bernoulli(p)	p chance of success	0 or 1	$p^a(1-p)^{1-a}$
Binomial(n,p)	p chance of success n number of trial	0 to n	$\binom{n}{a}(1-p)^a p^{n-a}$
Geometric(p)	p chance of success	0 to ∞	$(1-p)^a p$

Other distributions

V · T · E	Probability distributions (list)	[hide]
Discrete univariate	with finite support	Benford · Bernoulli · Beta-binomial · **Binomial** · Categorical · Hypergeometric (Negative) · Poisson binomial · Rademacher · Soliton · Discrete uniform · Zipf · Zipf–Mandelbrot
	with infinite support	Beta negative binomial · Borel · Conway–Maxwell–Poisson · Discrete phase-type · Delaporte · Extended negative binomial · Flory–Schulz · Gauss–Kuzmin · Geometric · Logarithmic · Mixed Poisson · Negative binomial · Panjer · Parabolic fractal · Poisson · Skellam · Yule–Simon · Zeta

Tips:

Select distributions that fits, and use the predefined pmf/cdf.

Quiz 2

Which distribution would you use for the following random variable:

1. A number of tries until you win an item from a lootbox?
2. Dice face when throwing a weighted dice?
3. The number of student passing this class?

Week 4 Homework

- Week 4 Homework is OUT now.
- Deadline: **MONDAY SEP-2.** **> NO LATE SUBMISSION <**
- Solution will be posted on Tuesday.
- Required assignment: Problem 5 and 7.

Finish for the day!